

Memo: Movie Recommendation Model

To: VP of Product

What was built: The current method for predicting movie ratings for users is to guess that all users rate all movies the mean. The model built predicts both how likely a user is to rate movies well and how likely a film is to be rated well. This can be used to find patterns in user behaviour and group users into groups of similar taste.

How Much Better is it: Two models were made- one that just replaced all unknown values with the mean and one that learned the differences between users. The simpler model had an average error of 0.9597, which means that on average it would be about one star off. The more complicated model had an average error of 0.8728, which is XXXX% better.

What the Model Says about the Audience: The largest difference between users appears to be between users who like watching movies with cultural relevance and users who like watching more generic movies. These movies tend to be watched and re-watched, while movies that were big at the time and then faded are less important. As a licensing call, this means we should try to get more older, more famous movies and focus less on big franchises and generic titles.

Two Limitations of the Model:

1. Some individual users received strange suggestions for their tastes. This could either happen if the user had one outlier the model fixed on to or the model would just recommend universally popular movies.
2. Some hidden patterns were vague. One hidden pattern found seemed to be a split between genres, but had some strange movies in it that don't apply (for example O Brother Where Art Thou being in the sci-fi/thriller category).

If We Had Two More Weeks: The largest improvements to this model can likely be made into focusing more on a user's specific patterns than the films as a whole. This would involve giving a user's ratings more weight than just how liked is a movie. With this improvement, StreamFlix could build a recommendation system that provides genuinely unique and personalized recommendations.

Reference: F. Maxwell Harper and Joseph A. Konstan. 2015. The MovieLens Datasets: History and Context. ACM Transactions on Interactive Intelligent Systems (TiiS) 5, 4: 19:1–19:19.
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